

A Formal Ontology and Automated Triage Framework for Theoretical Physics

The Epistemological Crisis in Scientific Classification

The contemporary landscape of theoretical physics is characterized by an unprecedented acceleration in the volume of published literature. With estimates suggesting that over 2.5 million scholarly articles are published annually across the scientific spectrum, the sheer magnitude of data presents a formidable challenge to knowledge retrieval, synthesis, and interdisciplinary discovery. Researchers are consistently inundated by a flood of information that surpasses human cognitive capacity for manual processing, necessitating advanced automated systems to parse, categorize, and route literature effectively. However, the foundational systems currently employed to classify physics research are structurally inadequate for the demands of modern automated triage and large language model (LLM) processing.

Historically, the classification of physics literature relied on rigid, manual categorizations. The Physics and Astronomy Classification Scheme (PACS), developed by the American Institute of Physics, served as the standard for decades. As the discipline evolved, PACS was recognized as excessively ad hoc and was subsequently replaced in 2016 by the Physics Subject Headings (PhySH), developed by the American Physical Society. PhySH introduced a two-dimensional faceted hierarchical vocabulary, separating professional "Disciplines" (such as Condensed Matter or Nuclear Physics) from conceptual "Facets" (such as Research Areas, Physical Systems, and Techniques). Concurrently, the arXiv repository, the primary preprint server for the physics community, utilizes a taxonomy based on broad sociological and historical boundaries, funneling papers into categories such as High Energy Physics - Theory (hep-th), Strongly Correlated Electrons (cond-mat.str-el), or Mathematical Physics (math-ph).

While these systems represented significant advancements in bibliographic organization, they remain fundamentally topical and sociological rather than ontological. They classify manuscripts based on the sub-discipline to which the author belongs or the general subject matter, rather than extracting the precise mathematical claims the theory makes regarding the nature of physical reality. This sociological sorting creates artificial barriers. For example, the mathematical architecture of the renormalization group applies equally to high-energy particle physics and condensed matter statistical mechanics, yet topical sorting segregates these applications, obscuring underlying theoretical symmetries.

Furthermore, as the scientific community increasingly turns to LLMs for automated text classification and literature triage, the limitations of topical taxonomies become acute. LLMs require logical rigor, mutual exclusivity where necessary, and precise semantic boundaries to avoid hallucination and misclassification. Topical categories often overlap ambiguously, lacking the strict Boolean differentiators required for algorithmic processing. To resolve these limitations and support the next generation of LLM-driven research agents, this report formalizes a unified, ontological tagging framework for theoretical physics.

Architectural Paradigm: Decoupling Ontology,

Epistemology, and Scale

The proposed framework reconstructs the classification of theoretical physics into a tripartite, multi-dimensional schema. Departing from traditional topical silos, this system extracts the foundational, operational, and methodological characteristics of the research. It mandates the strict separation of *what the theory claims about reality* (the ontology) from *how the theory is validated* (the epistemology or method) and *the dimensional level at which it operates* (the scale).

This decoupled architecture is divided into three orthogonal dimensions:

- 1. **Content Domains (Ontological/Mathematical):** Spanning through, these are mutually overlapping but conceptually distinct axes describing the fundamental nature of reality as posited by the text. They isolate claims regarding geometry, primitives, emergence, dynamics, information, and measurement.
- 2. **Method Flags (Implementation/Validation):** These labels classify the epistemological approach of the paper, identifying how the theoretical framework is instantiated, mathematically simulated, or empirically tested in physical or ecological systems.
- 3. **Optional Scale Modifiers:** These modifiers capture the dimensional or systemic level at which the theoretical framework primarily operates, providing critical context for cross-scale physical integration.

By combining these dimensions, an automated system can assign a highly descriptive, combinatorial tag to any manuscript. For example, a paper detailing a novel numerical simulation of a topological phase would be tagged as [Method-Comp]. This granularity facilitates highly specific knowledge retrieval and identifies canonical bridge patterns between seemingly disparate subfields.

Classification Dimension	Function within the Framework	Examples of Output Tags
Content Domains	Defines the explicit ontological and mathematical claims of the theory regarding the structure of the universe.	Symmetries, Primitives, Kinematics, Information Mechanics, Measurement
Method Flags	Identifies the empirical, computational, or operational implementation of the theory.	Simulation (Comp), Experiment (Exp), Living Ecosystems (Eco)
Scale Modifiers	Specifies the physical or systemic dimensional scale of the theoretical model.	Microscopic, Mesoscopic, Macroscopic, Multi-scale

Content Domains: Ontological and Mathematical Axes

The Content Domains (I–VI) constitute the core of the framework. They are designed to extract the deep theoretical commitments of a manuscript. The extraction process is governed by positive criteria defining what must be present, specific trigger keywords, and, crucially for LLM implementation, strict Boolean sorting prompts combined with negative exclusion rules to prevent false-positive classifications.

: Symmetries & Topologies

Philosophical and Physical Context: Domain-I captures the global geometric and topological

constraints that dictate the behavior of a physical system. In modern theoretical physics, particularly following the development of gauge theories and the classification of topological phases of matter, symmetry and topology are not merely descriptive auxiliary tools; they act as the fundamental organizing principles of reality. This domain encompasses research into Lie groups, fiber bundles, and conformal invariances that enforce strict constraints on allowable dynamics. It also captures the classification of exotic states of matter that exist beyond the traditional Landau symmetry-breaking paradigm, such as quantum spin liquids, fractional quantum Hall states, and systems exhibiting anyonic statistics or Majorana zero modes.

Positive Extraction Criteria: The text must identify the global or local geometrical structure, the mathematical invariants, or the topological features that constrain all dynamics within the proposed system. Furthermore, the text must define or analyze symmetry groups, spatial manifolds, gauge structures, or conserved quantities functioning as a background or organizing structure, rather than just employing them as secondary computational conveniences inside a specific phenomenological model.

Keywords and Triggers: Group theory, Lie groups, E_8 , gauge invariance, fiber bundles, spatial/space-time manifolds, topological phases, conservation laws, boundary conditions, anyons, Chern numbers, topological entanglement entropy.

Strict Boolean Sorting Prompt:

Does this text specify or analyze the global geometry, symmetry group, or topological/invariant constraint that all allowed states and dynamics must respect?

Negative Exclusion Cues: The automated system must *not* apply this tag if symmetry appears only as a passing remark or a simple discrete symmetry in an otherwise purely algorithmic model. Similarly, the system must exclude texts where "topology" is mentioned strictly metaphorically (e.g., "the topology of a computer network" or "the topology of ideas") without underlying mathematical geometric structure.

: Primitives & Irreducibles

Philosophical and Physical Context: Domain-II isolates the "Primitive Ontology" of the theory. Within the philosophy of physics, the primitive ontology approach asserts that any fundamental physical theory must possess a well-defined architecture whose foundation contains primitive variables representing the basic, irreducible constituents of matter in three-dimensional space or four-dimensional spacetime. This sharply contrasts with the nomological properties of a theory. For example, in many quantum interpretations, the wavefunction is viewed as nomological—it dictates the motion of the primitives but does not constitute physical matter itself. Domain-II identifies these fundamental building blocks—whether they are conceptualized as Bohmian point particles, strings, branes, continuous matter density fields, or discrete stochastic flashes (as in the GRWf approach)—that cannot be further decomposed within the theory's own framework.

Positive Extraction Criteria: The manuscript must propose or deeply analyze the fundamental building blocks (whether physical entities or pure mathematical structures) that cannot be decomposed. The text must explicitly specify "what exists" at the lowest level of reality, establishing a primitive ontology, fundamental basis elements, elementary nodes, or ontic entities.

Keywords and Triggers: Primitive ontology, local beables, fundamental particles, strings, branes, qubits as ontic units, prime number encodings, basis vectors, zero-dimensional charges, elementary events, nodes in a fundamental graph, matter density ontology.

Strict Boolean Sorting Prompt:

Does this text define what the physical or mathematical universe is ultimately made of at its lowest indivisible level within the theory?

Negative Exclusion Cues: The system must reject classification under Domain-II if the text only catalogs *effective* degrees of freedom in a phenomenological model without explicitly claiming fundamental or irreducible status. Furthermore, texts discussing “emergent quasiparticles” (such as phonons or magnons) must be excluded from this domain if they clearly treat these particles as derived from a deeper physical substrate that is not the primary subject of the analysis.

: Emergence & Partitioning

Philosophical and Physical Context: Domain-III addresses the mechanisms by which complex, macroscopic, fragmented, or lower-symmetry phenomena arise from unified, microscopic, or higher-symmetry substrates. It captures the theoretical essence of phase transitions, spontaneous symmetry breaking, and the mathematical framework of the renormalization group (RG) flow. The RG framework is particularly critical to this domain, providing the formal mathematical apparatus to investigate how physical laws and effective parameters (such as coupling constants and mass) evolve as the scale of observation varies, thereby explaining self-similarity and the emergence of macroscopic order from microscopic quantum fluctuations. This domain also encompasses cosmological emergence, such as inflationary epoch splits, the Higgs mechanism, and the emergence of classical spacetime from quantum entanglement.

Positive Extraction Criteria: The literature must explain *how* a unified, high-dimensional, or high-energy configuration transforms into a fragmented, lower-symmetry, or lower-dimensional observable regime. The text is required to provide a concrete narrative or mathematical mechanism for symmetry breaking, phase transitions, dimensional reduction, or epochal splits.

Keywords and Triggers: Symmetry breaking, phase transitions, dimensional reduction, renormalization group flows, running coupling constants, cosmological epochs, Higgs mechanism, confinement–deconfinement, emergence of space/time, emergence of classicality from a unified state.

Strict Boolean Sorting Prompt:

Does this text describe a concrete mechanism or process by which a more unified or higher-symmetry structure transforms into distinct, separated phases, forces, or fields?

Negative Exclusion Cues: Automated triage engines must reject texts that merely state that “X is emergent” without providing the mathematical or structural mechanism explaining *how* it emerges. Additionally, the system must exclude literature that uses the term “emergence” in a purely philosophical, sociological, or metaphorical sense, lacking a rigorous structural or dynamical physical model.

: Operators & Kinematics

Philosophical and Physical Context: Domain-IV identifies the generators of evolution and the rules of interaction within a theory. While Domain-I defines the geometric arena and Domain-II defines the irreducible actors, Domain-IV defines the script. This domain encompasses the formal mathematical structures that update states and drive evolution, such as Hamiltonians, Lagrangians, actions, and explicit equations of motion. It includes advanced kinematic frameworks, such as unified Lie-algebraic formulations, Foldy-Wouthuysen transformations utilized for block-diagonalizing relativistic Hamiltonians, and explicit operator constructions for

arbitrary spin fields. The focus remains strictly on the mathematical operators that drive dynamical change over a parameterized variable, typically time.

Positive Extraction Criteria: The text must define or analyze the generators of evolution and interaction: equations of motion, action functionals, Lagrangians, Hamiltonians, or other specific operators that update physical states. It must specify explicit kinematic or kinetic laws defining how entities move, accelerate, interact, or exchange physical quantities under the influence of these operators.

Keywords and Triggers: Lagrangian, Hamiltonian, action functional, Euler–Lagrange equations, Schrödinger equation, Heisenberg equation, field equations, force laws, propagators, kinetic terms, Lie-algebraic generators, Foldy-Wouthuysen transformation.

Strict Boolean Sorting Prompt:

Does this text provide or modify explicit evolution or interaction rules (e.g., an action, generator, or equation of motion) that determine how states change?

Negative Exclusion Cues: The system must not tag texts that discuss symmetry or conservation at a purely structural level without introducing or modifying specific dynamics; such texts belong exclusively in Domain-I. Furthermore, the system must exclude papers focusing solely on numerical methods, computer solver algorithms, or discretizations of already-known standard equations without proposing or analyzing new physical operators.

: Information Mechanics & Entropy

Philosophical and Physical Context: Domain-V tracks the flow, storage, and degradation of information within physical and mathematical systems. It represents the modern intersection of thermodynamics, quantum mechanics, and information theory. Key theoretical constructs within this domain include the holographic principle—which posits that the entire information content of a volume of space is encoded on its lower-dimensional boundary, resolving paradoxes related to black hole thermodynamics and the Bekenstein bound. Furthermore, Domain-V encompasses research into entanglement entropy, quantum error-correcting codes, and the mapping of physical space-time geometry to tensor networks and quantum communication channels.

Positive Extraction Criteria: The literature must track where physical “data” (such as microstates, classical bits, qubits, or records) reside, how they propagate, and how entropy or information measures change under system dynamics. The text must develop or apply explicit information-theoretic, entropic, or coding-theoretic structures to evaluate physical or mathematical systems.

Keywords and Triggers: Quantum information theory, entanglement entropy, mutual information, channel capacity, error-correcting codes, holographic principle, AdS/CFT correspondence, black hole thermodynamics, microstates, entanglement networks, Landauer bounds.

Strict Boolean Sorting Prompt:

Does this text explicitly model or calculate the storage, transfer, or degradation of information (classical or quantum) in terms of microstates, channels, or entropic quantities?

Negative Exclusion Cues: The triage system must strictly exclude texts that mention “entropy” or “information” only as a general metaphor (e.g., “the informational universe” or “the entropy of modern society”) without precise mathematical definitions or calculations. Additionally, the system must exclude literature that uses entropy purely as a macroscopic thermodynamic state function in a standard engineering context, devoid of any discussion regarding microstates, probability distributions, or information-theoretic channels.

: State Resolution & Measurement

Philosophical and Physical Context: Domain-VI is dedicated to the measurement problem and the transition from the quantum realm to classical reality. It isolates texts that attempt to explain or formalize how superposed quantum possibilities resolve into definite, macroscopic observable outcomes. This includes theoretical mechanisms for objective wavefunction collapse (such as the Ghirardi-Rimini-Weber or GRW models), environment-induced decoherence, and the selection of the pointer basis. It rigorously separates the precise *process* of state resolution from the general kinematics of standard unitary evolution captured in Domain-IV.

Positive Extraction Criteria: The text must directly address the measurement problem or the transition from superposed quantum possibilities to single classical outcomes. It must propose or deeply analyze specific mechanisms, mathematical conditions, or formalisms for wavefunction collapse, decoherence, or other superposition-resolution processes.

Keywords and Triggers: Wavefunction collapse, objective collapse models, decoherence, pointer basis, environment-induced superselection, observer effect, quantum interpretations (Copenhagen, Many Worlds, Bohmian, GRW, etc.), quantum–classical boundary.

Strict Boolean Sorting Prompt:

Does this text propose or analyze a specific condition, mechanism, or equation that turns quantum superpositions into definite, classically measurable outcomes?

Negative Exclusion Cues: The system must not tag texts that discuss measurement only at an operational or procedural level (e.g., “we then measure the photon spin using a detector”) without engaging theoretically with the superposition-to-outcome transition. The system must also exclude purely philosophical or historical discourse that lacks any link to a specific physical mechanism, mathematical formalism, or ontological condition for outcome definiteness.

Method Flags: Decoupling Implementation from Ontology

A significant limitation of legacy taxonomies is the conflation of the theoretical ontology with the method used to validate it. For instance, a paper performing a Monte Carlo simulation of string theory might struggle to be accurately tagged in both hep-th (High Energy Physics - Theory) and physics.comp-ph (Computational Physics) without confusing the primary theoretical contribution with the computational tool utilized. To resolve this, the framework recasts implementation and validation strategies as orthogonal *Method Flags* that can be appended to any Content Domain.

[Method-Comp] : Computational / Algorithmic Implementation

Context and Extraction: This flag identifies research that translates theoretical physics into executable algorithms, data structures, or numerical simulations. It encompasses lattice models, Monte Carlo simulations, and numerical contour integrations utilized to approximate or visualize complex, multi-scale dynamics.

Keywords and Triggers: Computational physics, numerical simulation, Monte Carlo, lattice models, agent-based models, algorithm design, code repositories, data architectures, software frameworks.

Strict Boolean Prompt:

Does this text implement, simulate, or algorithmically explore a theory or model to test, approximate, or visualize its behavior through computer code?

[Method-Exp] : Experimental / Empirical Validation

Context and Extraction: This flag is applied to literature focused on the physical, empirical testing of theoretical predictions. It encompasses the design of laboratory apparatus, measurement protocols, empirical benchmark experiments, and direct data analysis linked to validating a theoretical framework.

Keywords and Triggers: Experimental test, measurement protocol, apparatus design, laboratory implementation, empirical validation, benchmark experiments, observational campaigns.

Strict Boolean Prompt:

Does this text describe an actual or proposed physical experiment, laboratory setup, or observational protocol to empirically test a theoretical claim?

[Method-Eco] : Living Ecosystems & Operational Embodiments

Context and Extraction: [Method-Eco] represents a highly specialized yet increasingly prominent methodological approach where physical and mathematical structures are embedded within adaptive, living, or socio-technical systems. This encompasses concepts such as "living mathematics ecosystems," bio-inspired physics, complex adaptive systems, and evolutionary agent-based models. It applies when researchers utilize biological, ecological, or sociotechnical frameworks to instantiate or test physical theories—for instance, using evolutionary population dynamics to model phase transitions, or studying how adaptive networks amplify information-mechanical principles over time.

Keywords and Triggers: Living mathematics ecosystems, artificial life, evolutionary simulations, adaptive multi-agent systems, bio-inspired computation, ecosystem experiments, complex adaptive systems.

Strict Boolean Prompt:

Does this text instantiate the physical or mathematical theory within an evolving, biological, or adaptive ecosystem (real or simulated) to study its behavior as a complex living system over time?

Optional Scale Modifiers: Resolving Dimensional Hierarchy

Physical phenomena frequently exhibit profoundly different behaviors depending on the scale of observation. Scale modifiers provide lightweight, non-exclusive labels that anchor the primary descriptive level of the paper. These modifiers do not alter the fundamental meaning of the Content Domains; rather, they provide necessary context regarding the physical extent of the system under study.

- : Pertains to the fundamental, microscopic, quantum, or discrete particle levels. It includes Planck-scale phenomena, individual atomic interactions, and discrete event-based microscale algorithms.
- : Describes intermediate, localized collective scales. It captures complex networks, mesoscopic materials, localized biological systems, and intermediate statistical ensembles described by Boltzmann-type kinetic equations.
- : Captures large-scale, continuous, or systemic phenomena. It applies to cosmological

structures, astrophysical events, planetary dynamics, and large socio-technical infrastructures described by partial differential equations.

- : Applied when a paper explicitly models the interaction, coupling, or flow between multiple scales. It is crucial for research involving the renormalization group, homogenization methods, or concurrent multi-scale modeling linking atomistic behavior to continuum mechanics.

Priority Tagging and Canonical Bridge Patterns

Because high-level scientific research is inherently intersectional, it is rare for a paper to fall strictly within a single ontological domain. To accurately encode multi-domain papers and represent the conceptual topography of modern physics, the framework utilizes **Primary** and **Secondary** priority markers.

- `` indicates that the central theoretical contribution and focal point of the manuscript lie within Domain X.
- `` indicates that Domain Y is substantially engaged or modified by the research, though it remains subordinate to the primary focus.

This hierarchical tagging structure inherently reveals "Canonical Bridge Patterns," allowing automated systems to map how different subfields interact and drive theoretical innovation.

Canonical Bridge Pattern	Taxonomic Output Designation	Conceptual Description of the Research
Emergent Geometry from Entanglement	 ``	Models demonstrating how macroscopic spatial manifolds and geometries (Domain I) emerge (Domain III) directly from underlying quantum information networks or entanglement entropy (Domain V).
Simulated Objective Collapse Models	 `` [Method-Comp]	Papers analyzing modifications to the standard Schrödinger equation (Domain IV) to force objective wavefunction collapse and state resolution (Domain VI), validated via numerical simulation (Method-Comp).
Topological Hamiltonian Synthesis	 `` [Method-Comp]	Research defining explicit kinematic generators and equations of motion (Domain IV) that yield robust topological invariants and protected phases (Domain I), tested algorithmically.
Living Mathematics in Ecosystems	 `` [Method-Eco]	Studies tracking how information mechanics and entropy (Domain V) drive macroscopic emergence

Canonical Bridge Pattern	Taxonomic Output Designation	Conceptual Description of the Research
		(Domain III) within an adaptive, bio-inspired multi-agent system (Method-Eco).

LLM Deployment and Prompt Engineering for Automated Triage

The operationalization of this formal framework relies heavily on deploying Large Language Models (LLMs) to automate the triage and tagging of scholarly texts. Recent studies indicate that advanced LLMs (such as Llama 3.1, GPT-4, and Gemini 2.5) have demonstrated remarkable proficiency in text classification, achieving accuracies up to 0.82 on complex hierarchical classification datasets like the Field of Research Classification (FoRC), effectively rivaling and often surpassing traditional supervised learning algorithms such as BERT. However, the deployment of LLMs for automated scientific triage is not without significant challenges. A primary obstacle is "prompt brittleness," a phenomenon wherein minor semantic shifts in the user instruction or input formatting can lead to drastic variations in classification accuracy, false positives, or model hallucinations. To deploy this ontology reliably at scale, the triage system must utilize structured, agentic workflows rather than relying on simple, unstructured zero-shot inquiries.

Integrating Strict Boolean Prompts and Negative Cues

The explicit Boolean prompts detailed in the Content Domains section act as the primary filters for the LLM. By framing the classification task as a strict yes/no determination based on clearly defined structural attributes, the model's action space is tightly constrained. Crucially, to prevent "over-triage"—where models assign tags based on passing metaphorical references rather than deep structural mathematical engagement—the framework mandates the use of negative extraction cues. The inclusion of these negative cues establishes rigid boundary conditions. For instance, if an LLM is evaluating a paper for (Symmetries & Topologies), the system prompt must explicitly incorporate the directive: *"Do NOT tag if topology is mentioned only metaphorically (e.g., 'topology of ideas') with no supporting mathematical structure."* Empirical research underscores that constraining the LLM with highly specific exclusion criteria drastically reduces hallucination rates and improves the signal-to-noise ratio during complex text annotation tasks.

Few-Shot and Chain-of-Thought (CoT) Optimization

To maximize classification accuracy, the automated triage architecture should employ advanced prompting methodologies, specifically Few-Shot Learning and Chain-of-Thought (CoT) prompting.

- **Few-Shot Prompting:** Providing the LLM with a limited number of expertly annotated abstracts demonstrating the nuanced difference between domains—for example, distinguishing the purely structural claims of Domain-I from the dynamic generators of Domain-IV—calibrates the model's contextual understanding. This exemplar-driven approach establishes a much higher performance ceiling than zero-shot classification.

- **Chain-of-Thought (CoT):** Instructing the model to generate step-by-step reasoning prior to outputting the final tags forces the LLM to explicitly match the paper's contents against the positive criteria and the negative exclusion cues before finalizing the classification. Studies indicate that CoT prompting significantly enhances the ability of models to engage in structured, multi-step reasoning, particularly for nuanced outputs.

By orchestrating these LLM capabilities around the rigid ontological axes of Domains I-VI, alongside the Method Flags and Scale Modifiers, research institutions can automate the ingestion, mapping, and routing of theoretical physics literature with near-human reliability and unprecedented scalability.

Conclusions

The formal framework detailed in this report represents a necessary paradigm shift in the classification of theoretical physics literature. By transitioning away from the topical, sociologically-driven silos of traditional taxonomies—such as PACS and the current arXiv categorizations—to a rigorous, ontological extraction model, this schema accurately captures the fundamental mathematical and physical claims of modern research.

The strategic decoupling of Content Domains (what the theory says reality is like) from Method Flags (how the theory is validated) and Scale Modifiers (the dimensional scope) resolves the persistent problem of multidimensional research being forced into singular, reductive categories. Furthermore, the explicit design of the taxonomy—utilizing strict Boolean prompts, positive extraction criteria, and vital negative boundary cues—makes it natively optimized for deployment via advanced Large Language Models. Implementing this structured framework within an agentic LLM workflow will allow institutions to automate the triage of the rapidly expanding scientific corpus, mitigate the risks of model hallucination, uncover hidden canonical bridge patterns, and ultimately accelerate cross-disciplinary scientific discovery.

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